

```
In [1]: import pandas as pd
```

```
In [2]: data=pd.read_csv("Electricity.csv")
data
```

```
Out[2]:
```

	Time_in_mint	HolidayFlag	DayOfWeek	WeekOfYear	Day	Month	Year	PeriodOfDay	Fo
0	0	0	1	44	1	12	2019	0	
1	30	0	1	44	1	12	2019	1	
2	60	0	1	44	1	12	2019	2	
3	90	0	1	44	1	12	2019	3	
4	120	0	1	44	1	12	2019	4	
...	...	...	...	...	...	...	...	...	...
94	1410	0	2	44	2	12	2019	46	
95	1440	0	2	44	2	11	2019	47	
96	1470	0	3	44	3	12	2019	0	
97	1500	0	3	44	3	11	2019	1	
98	1530	0	1	21	22	6	2020	27	

99 rows × 17 columns



```
In [3]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99 entries, 0 to 98
Data columns (total 17 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Time_in_mint                          99 non-null     int64
1   HolidayFlag                           99 non-null     int64
2   DayOfWeek                             99 non-null     int64
3   WeekOfYear                            99 non-null     int64
4   Day                                    99 non-null     int64
5   Month                                 99 non-null     int64
6   Year                                  99 non-null     int64
7   PeriodOfDay                           99 non-null     int64
8   ForecastWindProduction                99 non-null     float64
9   SystemLoadEA                          99 non-null     float64
10  SMPEA                                 99 non-null     float64
11  ORKTemperature                         96 non-null     float64
12  ORKWindspeed                          96 non-null     float64
13  CO2Intensity                           99 non-null     float64
14  ActualWindProduction                  95 non-null     float64
15  SystemLoadEP2                         99 non-null     float64
16  SMPEA2                                99 non-null     float64
dtypes: float64(9), int64(8)
memory usage: 13.3 KB
```

```
In [4]: data.isnull().sum()
```

```
Out[4]: Time_in_mint      0
HolidayFlag             0
DayOfWeek               0
WeekOfYear              0
Day                    0
Month                  0
Year                  0
PeriodOfDay            0
ForecastWindProduction  0
SystemLoadEA           0
SMPEA                  0
ORKTemperature          3
ORKWindspeed            3
CO2Intensity            0
ActualWindProduction    4
SystemLoadEP2           0
SMPEA2                  0
dtype: int64
```

```
In [5]: data=data.dropna()
```

```
In [6]: data
```

```
Out[6]:
```

	Time_in_mint	HolidayFlag	DayOfWeek	WeekOfYear	Day	Month	Year	PeriodOfDay	Fo
0	0	0	1	44	1	12	2019	0	
1	30	0	1	44	1	12	2019	1	
2	60	0	1	44	1	12	2019	2	
3	90	0	1	44	1	12	2019	3	
4	120	0	1	44	1	12	2019	4	
...	...	...	...	...	...	...	...	...	...
94	1410	0	2	44	2	12	2019	46	
95	1440	0	2	44	2	11	2019	47	
96	1470	0	3	44	3	12	2019	0	
97	1500	0	3	44	3	11	2019	1	
98	1530	0	1	21	22	6	2020	27	

93 rows × 17 columns



```
In [7]: from sklearn.preprocessing import Normalizer
scaler=Normalizer()
scaled_data=scaler.fit_transform(data)
scaled_df=pd.DataFrame(scaled_data,columns=data.columns)
scaled_df
```

0	0.000000	0.0	0.000196	0.008606	0.000196	0.002347	0.394913
1	0.006161	0.0	0.000205	0.009036	0.000205	0.002464	0.414614
2	0.012780	0.0	0.000213	0.009372	0.000213	0.002556	0.430063
3	0.019756	0.0	0.000220	0.009659	0.000220	0.002634	0.443195
4	0.026947	0.0	0.000225	0.009881	0.000225	0.002695	0.453384
...	...	...	...	...	...	...	...
88	0.223468	0.0	0.000317	0.006973	0.000317	0.001902	0.319988
89	0.236285	0.0	0.000328	0.007220	0.000328	0.001805	0.331292
90	0.255164	0.0	0.000521	0.007638	0.000521	0.002083	0.350460
91	0.271826	0.0	0.000544	0.007974	0.000544	0.001993	0.365878
92	0.226458	0.0	0.000148	0.003108	0.003256	0.000888	0.298984

93 rows × 17 columns

```
In [8]: from sklearn.preprocessing import StandardScaler
scaler=StandardScaler()
scaled_data=scaler.fit_transform(data)
scaled_df=pd.DataFrame(scaled_data,columns=data.columns)
scaled_df
```

```
Out[8]:
```

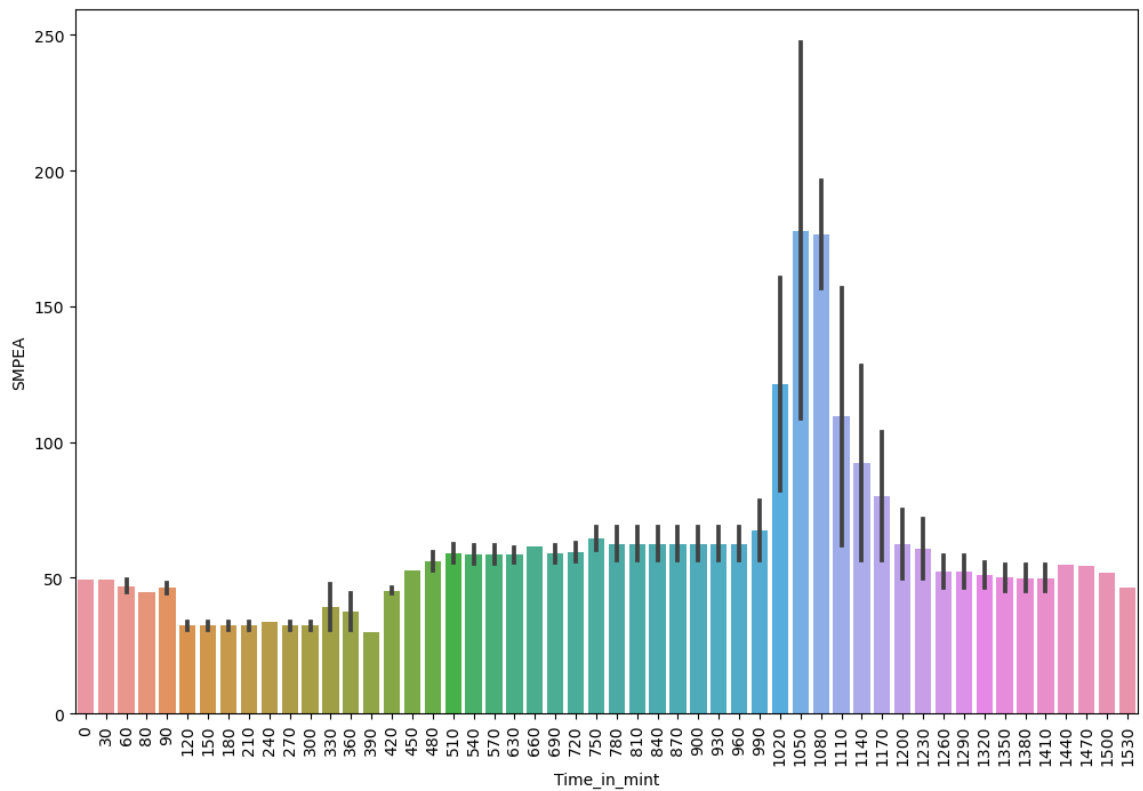
	Time_in_mint	HolidayFlag	DayOfWeek	WeekOfYear	Day	Month	Year	Peri
0	-1.740059	0.0	-0.953914	0.104257	-0.340402	0.195646	-0.104257	-
1	-1.671421	0.0	-0.953914	0.104257	-0.340402	0.195646	-0.104257	-
2	-1.602783	0.0	-0.953914	0.104257	-0.340402	0.195646	-0.104257	-
3	-1.534145	0.0	-0.953914	0.104257	-0.340402	0.195646	-0.104257	-
4	-1.465507	0.0	-0.953914	0.104257	-0.340402	0.195646	-0.104257	-
...	...	...	...	...	...	...	...	...
88	1.485926	0.0	0.894295	0.104257	0.118401	0.195646	-0.104257	-
89	1.554564	0.0	0.894295	0.104257	0.118401	-1.320613	-0.104257	-
90	1.623202	0.0	2.742504	0.104257	0.577203	0.195646	-0.104257	-
91	1.691840	0.0	2.742504	0.104257	0.577203	-1.320613	-0.104257	-
92	1.760478	0.0	-0.953914	-9.591663	9.294444	-8.901911	9.591663	-

93 rows × 17 columns

```
In [9]: import matplotlib.pyplot as plt
import seaborn as sn
%matplotlib inline
```

```
In [10]: plt.figure(figsize=(12,8))
sn.barplot(x='Time_in_min', y='SMPEA', data=data)
plt.xticks(rotation=90)
```

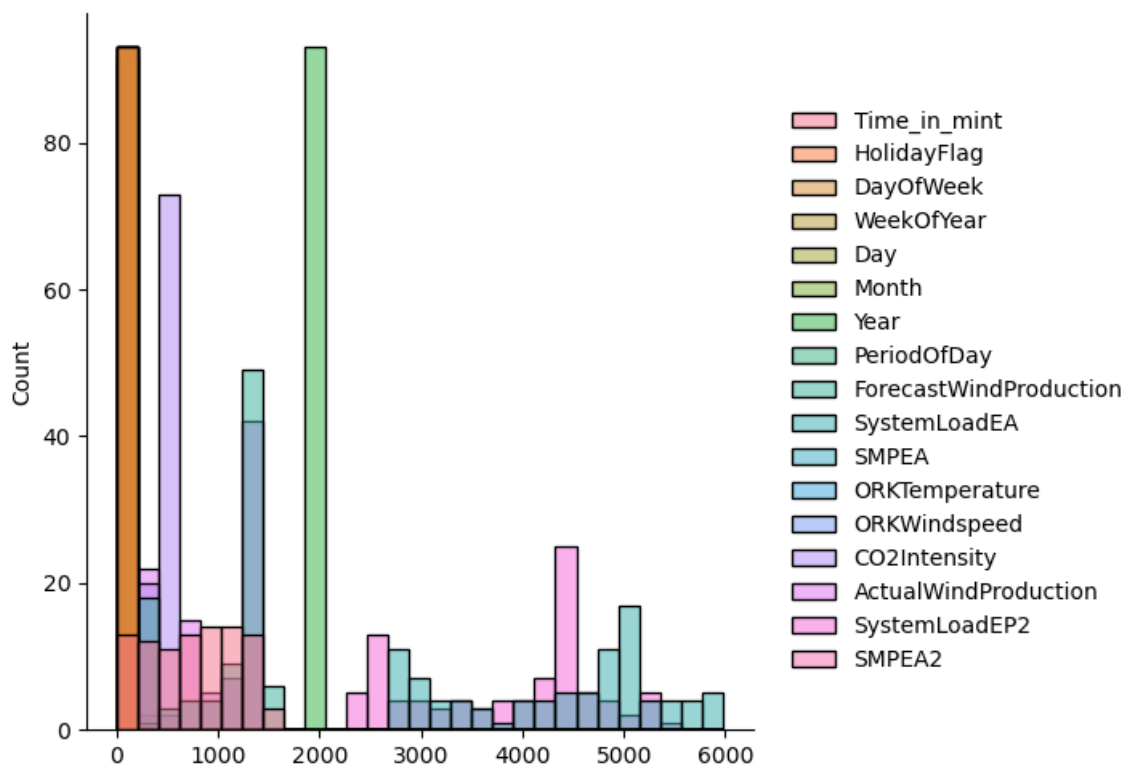
```
Out[10]: (array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
                17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
                34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
                51]),
          [Text(0, 0, '0'),
           Text(1, 0, '30'),
           Text(2, 0, '60'),
           Text(3, 0, '80'),
           Text(4, 0, '90'),
           Text(5, 0, '120'),
           Text(6, 0, '150'),
           Text(7, 0, '180'),
           Text(8, 0, '210'),
           Text(9, 0, '240'),
           Text(10, 0, '270'),
           Text(11, 0, '300'),
           Text(12, 0, '330'),
           Text(13, 0, '360'),
           Text(14, 0, '390'),
           Text(15, 0, '420'),
           Text(16, 0, '450'),
           Text(17, 0, '480'),
           Text(18, 0, '510'),
           Text(19, 0, '540'),
           Text(20, 0, '570'),
           Text(21, 0, '630'),
           Text(22, 0, '660'),
           Text(23, 0, '690'),
           Text(24, 0, '720'),
           Text(25, 0, '750'),
           Text(26, 0, '780'),
           Text(27, 0, '810'),
           Text(28, 0, '840'),
           Text(29, 0, '870'),
           Text(30, 0, '900'),
           Text(31, 0, '930'),
           Text(32, 0, '960'),
           Text(33, 0, '990'),
           Text(34, 0, '1020'),
           Text(35, 0, '1050'),
           Text(36, 0, '1080'),
           Text(37, 0, '1110'),
           Text(38, 0, '1140'),
           Text(39, 0, '1170'),
           Text(40, 0, '1200'),
           Text(41, 0, '1230'),
           Text(42, 0, '1260'),
           Text(43, 0, '1290'),
           Text(44, 0, '1320'),
           Text(45, 0, '1350'),
           Text(46, 0, '1380'),
           Text(47, 0, '1410'),
           Text(48, 0, '1440'),
           Text(49, 0, '1470'),
           Text(50, 0, '1500'),
           Text(51, 0, '1530')])
```



In [11]: `sn.displot(data)`

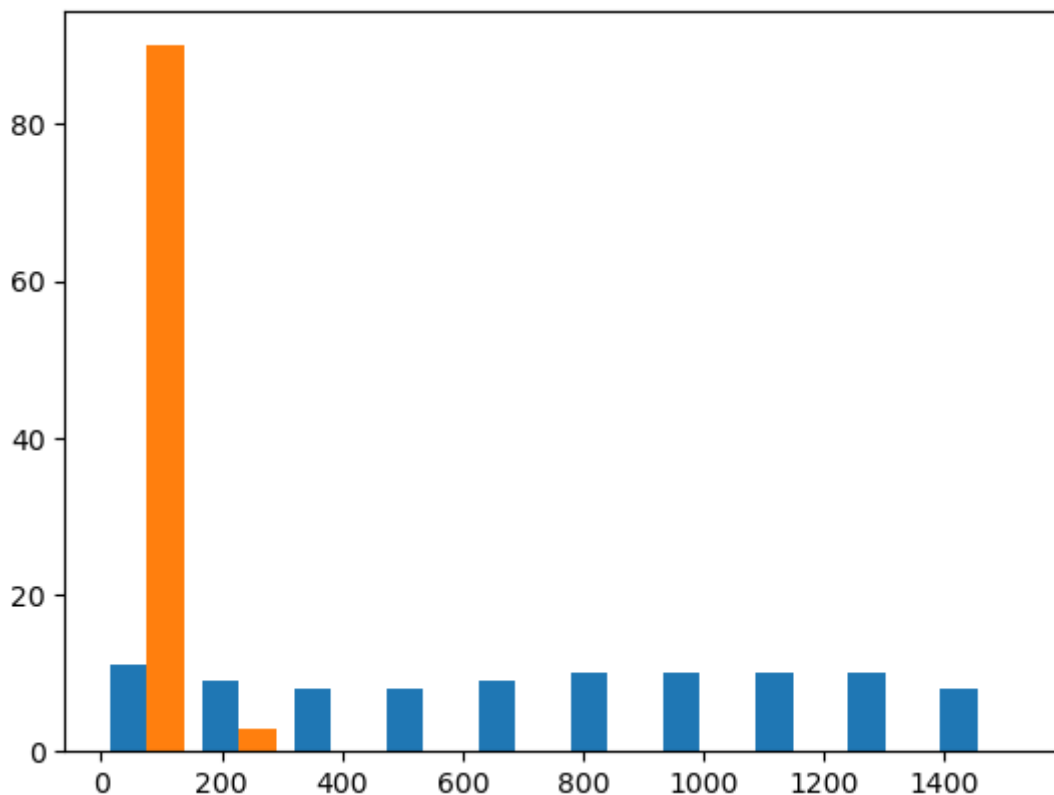
C:\Users\Praveena\anaconda3\Lib\site-packages\seaborn\axisgrid.py:118: Use
rWarning: The figure layout has changed to tight
self._figure.tight_layout(*args, **kwargs)

Out[11]: <seaborn.axisgrid.FacetGrid at 0x25a72c62a50>



```
In [12]: plt.hist(data[['Time_in_mint', 'SMPEA2']])
```

```
Out[12]: (array([[11.,  9.,  8.,  8.,  9., 10., 10., 10., 10.,  8.],
                [90.,  3.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.]]),
          array([  0., 153., 306., 459., 612., 765., 918., 1071., 1224.,
                1377., 1530.]),
          <a list of 2 BarContainer objects>)
```



```
In [13]: x = data[["Day", "Month", "ForecastWindProduction", "SystemLoadEA",
                  "SMPEA", "ORKTemperature", "ORKWindspeed", "CO2Intensity",
                  "ActualWindProduction", "SystemLoadEP2"]]
y = data["SMPEA2"]
from sklearn.model_selection import train_test_split
xtrain, xtest, ytrain, ytest = train_test_split(x, y, test_size=0.2, random
```

```
In [14]: from sklearn.ensemble import RandomForestRegressor
model = RandomForestRegressor()
model.fit(xtrain, ytrain)
```

```
Out[14]: RandomForestRegressor()
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

```
In [15]: y_pre=model.predict(xtest)
print(y_pre)
```

```
[54.6931 58.1459 14.4493 57.262  53.9093 54.2843 55.1036 56.8121 39.5279
 54.8399 53.9024 79.4474 47.3193 53.9395 45.6805 39.5715 58.6127 54.0664
 29.3103]
```

In []: